Skin grafting is a procedure in which healthy skin from a different region of the body is transplanted onto an area of absent or damaged skin. Skin grafts are used for partial-thickness burns, which damage the epidermis and dermis, and for full-thickness burns, which damage the epidermis, dermis, and hypodermis. (Superficial burns injure only epidermal tissue and can heal without skin grafts.) Skin grafting has greatly improved the long-term survival of burn victims. However, grafted skin cannot effectively dissipate heat, which places burn survivors at an increased risk of hyperthermia (abnormally high body temperature), a condition that can lead to heat-related medical emergencies such as heat stroke. The greater the grafted body surface area (BSA), the greater the risk for complications related to thermoregulation.

Heat acclimation (HA) is the process by which the body increases heat tolerance through adaptive changes in the sympathetic control of skin heat dissipation. In healthy individuals, HA can be induced by repeated exercise at elevated ambient temperature and humidity. A group of researchers tested whether skin graft recipients could also undergo HA. Skin graft recipients who were otherwise healthy and had at least 15% grafted BSA were recruited for the study, as were controls with no history of receiving grafts. The skin graft recipients were split into two groups, those with 15%–40% grafted BSA and those with >40% grafted BSA. For a period of seven days, subjects participated daily in 90 min of exercise at 50% maximal oxygen consumption in a room where the temperature was maintained at 40°C. Exercise modality, fluid intake, and frequency and duration of rest were strictly monitored.

On the days immediately preceding and following the HA regimen, the heat tolerance of participants was tested with a 90-min exercise regimen at a fixed heat production rate (Figure 1). Tolerance testing was halted before 90 min for subjects whose internal body temperature exceeded 40°C. The change in internal temperature of participants was calculated by subtracting their internal temperature before exercise from their internal temperature at exercise cessation (ie, either once temperature reached 40°C or at the 90-min endpoint).

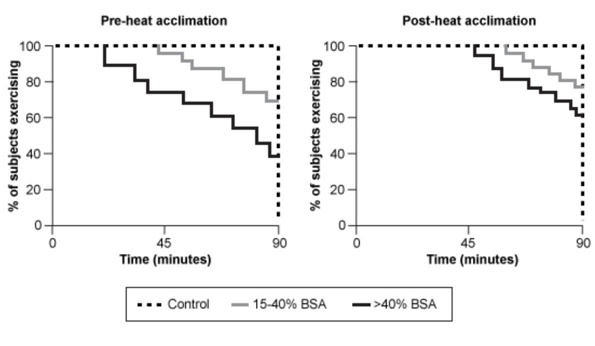


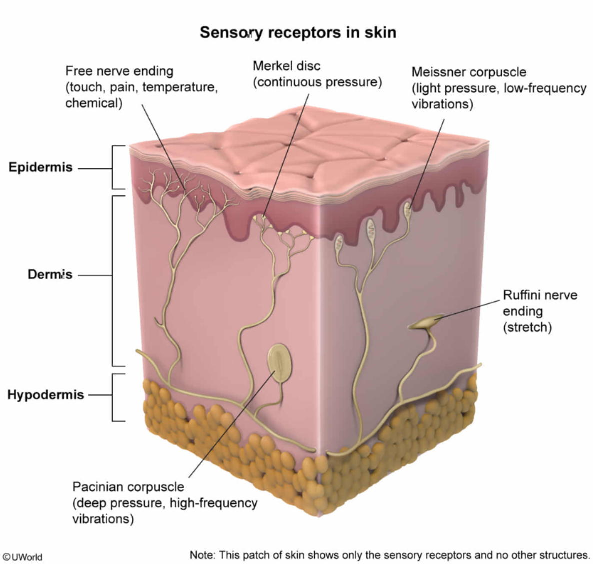
Figure 1 HA in controls and in skin graft recipients

Prior to receiving a skin graft, a patient with full-thickness burns on approximately 50% of their total BSA is at risk for all the following EXCEPT:

- A. extensive fluid loss. [28%]
- B. bacterial infection. [5%]
- ✔ C. increased sensitivity to touch. [59%]
- D. decreased abrasion protection. [6%]

Correct
58% answered correctly

Explanation:



The skin contains numerous sensory receptors that respond to external environmental stimuli. Free nerve endings react to thermal, chemical, mechanical, and potentially harmful stimuli and are found throughout the epidermis and dermis. Other sensory receptors in the epidermis (eg, Merkel discs) and dermis (eg, Meissner corpuscles, Ruffini endings, Pacinian corpuscles) detect various types of mechanical stimulation such as high- and low-frequency vibrations, light and heavy pressure, and stretch.

According to the passage, the damage associated with a full-thickness burn extends through the epidermis and dermis and into the subcutaneous layer (ie, hypodermis). Therefore, this type of burn involves injury to the epidermal and dermal sensory receptors, resulting in **impaired touch sensation**. The question asks what a patient with full-thickness burns is *not* at risk for; consequently, this patient would be at risk for decreased sensitivity to touch, not **increased sensitivity to touch**.

(Choice A) Intercellular tight junctions and lipids between epidermal cells form a waterproof barrier that limits the escape of water from the body. Sebum, an oily substance secreted onto the skin surface by sebaceous (oil) glands, also plays a minimal role in waterproofing the skin. A patient with full-thickness burns would be less able to retain fluids and electrolytes because the physical barrier that normally seals water inside the body (ie, the skin's outer layer) has been damaged or lost; therefore, this patient *is* at risk for extensive fluid loss.

this patient is at risk for extensive fluid loss.

(Choice B) The skin is part of the innate immune system, protecting the body from

pathogens by acting as a physical barrier that blocks their entry and housing immune cells that attack invading microbes. Full-thickness burns destroy these immune defenses and increase the individual's risk for infection; therefore, this patient *is* at risk for bacterial infection.

(Choice D) The skin is important in protection against abrasion. For example, calluses form on regions of the skin repeatedly subjected to friction. Because this individual has suffered damage to approximately 50% of their total BSA, the skin is no longer able to contribute as effectively to abrasion protection; therefore, this patient *is* at risk for decreased abrasion protection.

Educational objective:
The skin functions as a physical barrier to protect against abrasion and to prevent the loss of fluid from the body while simultaneously blocking the entry of pathogens. The skin also contains receptors that gather and respond to sensory information from the surrounding environment.

Time spent:0 sec
Subject:
Biology

QID:400932
System:
Skin and Immune Systems

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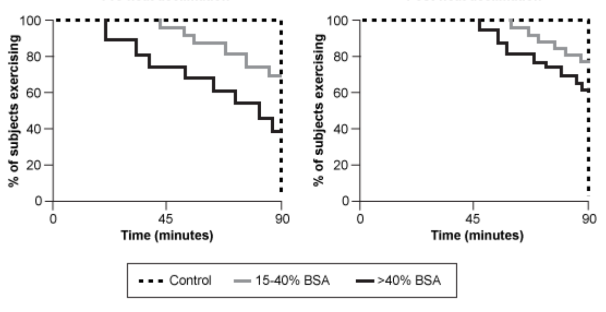


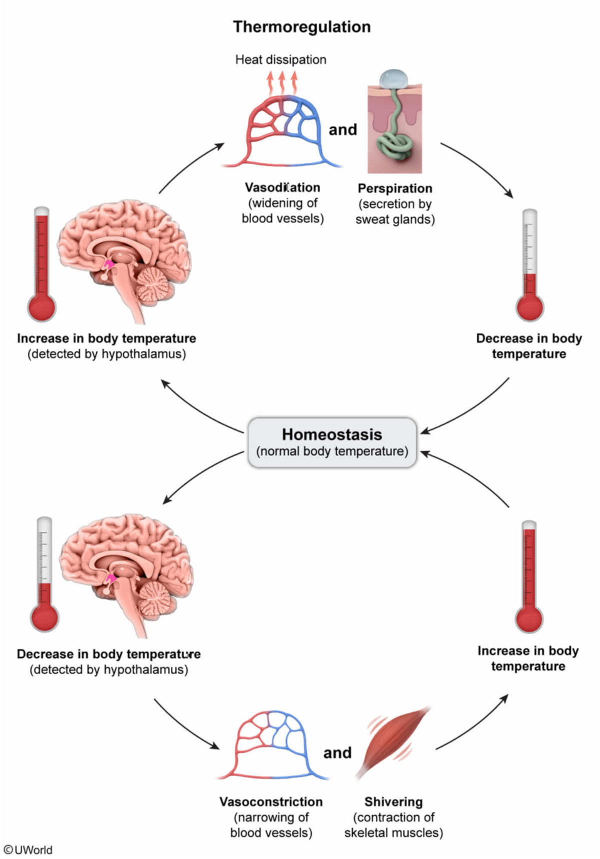
Figure 1 HA in controls and in skin graft recipients

What physiological change occurs in the skin of control subjects during HA exercise?

- ◯ A. Enhanced contraction of arteriole smooth muscle [11%]
- ✔ ◉ B. Enhanced relaxation of arteriole smooth muscle [48%]
- ◯ C. Enhanced contraction of arrector pili muscles due to parasympathetic activation [4%]
- ◯ D. Enhanced contraction of arrector pili muscles due to sympathetic activation [35%]

Correct
47% answered correctly

Explanation:



Thermoregulation, the maintenance of body temperature within the normal physiological range, is a major homeostatic function of skin. Thermoregulatory centers in the hypothalamus monitor body temperature by processing information from internal and surface (skin) thermoreceptors. When a significant change in body temperature is detected, the hypothalamic center coordinates the physiological responses necessary to reset the temperature back within the normal range.

When **body temperature** is **above normal** (as would occur during HA exercise), the following processes **cool the body** (ie, decrease body temperature):

- **Vasodilation (widening) of skin arterioles** increases blood flow to skin capillaries and maximizes heat loss through the skin. This occurs because blood, which is warmed in the body core, transfers heat to the environment when it passes through skin capillaries. Vasodilation occurs when the **smooth muscle surrounding the blood vessels relaxes**.

- **Sweat**, a hypotonic solution, is secreted onto the skin surface by sweat (sudoriferous) glands. Heat loss and subsequent cooling occur due to the evaporation of the water in the sweat, an endothermic process that absorbs heat from the body.

When **body temperature** is **below normal**, the following processes **warm the body** (ie, increase body temperature):

- **Vasoconstriction (narrowing) of skin arterioles** minimizes heat loss by diverting warm blood away from skin capillaries and toward blood vessels in the interior of the body. Vasoconstriction occurs when the smooth muscle surrounding the blood vessels contracts **(Choice A)**.

- **Shivering** generates heat through rhythmic involuntary contractions of skeletal muscle.

(Choices C and D) In this scenario, the controls are exercising in a hot environment, so the arrector pili muscles (attached to hair follicles in the skin) would not contract. However, in *cold* environments, *sympathetic* signaling causes contraction of the arrector pili muscles, which causes piloerection (ie, hairs standing upright). Piloerection, which impedes heat loss by trapping heat near the skin surface, is not an efficient means of heat retention in humans due to insufficient body hair.

Educational objective:
One major function of the skin is thermoregulation. Body temperature can be increased by vasoconstriction of skin arterioles, shivering, and piloerection. Body temperature can be decreased by vasodilation of skin arterioles and sweating.

Time spent:0 sec	**QID**:400933
Subject:	**System:**
Biology	Skin and Immune Systems